using python as a calculator

number

In [1]:
```
2+2
```
Out[1]: 4

In [2]:
```
50 - 5*6
```
Out[2]: 20

In [3]:
```
(50 - 5*6) / 4
```
Out[3]: 5.0

In [8]:
```
8 / 5  # division always returs to a floating-point number
```
Out[8]: 1.6

In [5]:
```
17 / 3  # classic division reutns to float
```
Out[5]: 5.666666666666667

In [6]:
```
17 // 3  #floor division discards the fractional part
```
Out[6]: 5

In [7]:
```
17 % 3   # the % operator returns the remainder of the division
```
Out[7]: 2

In [9]:
```
5 * 3 + 2  # floored quotient * divisor + remainder
```
Out[9]: 17

In [10]:
```
5 ** 2  #5 squared
```
Out[10]: 25

In [11]:
```
2 ** 7 # 2 to the power 7
```
Out[11]: 128

In [13]:
```
width = 20
height =5 * 9
width * height
```
Out[13]: 900

In [16]:
```
4 * 3.75 - 1
```

Out[16]: 14.0

In [17]:
```
tax = 12.5 / 100
price = 100.50
price * tax
```

Out[17]: 12.5625

In [18]:
```
price + _
```

Out[18]: 113.0625

In [19]:
```
round(_, 2)
```

Out[19]: 113.06

text

In [20]:
```
'spam eggs'
```

Out[20]: 'spam eggs'

In [22]:
```
"Paris rabbit got your back :)! yay!"
```

Out[22]: 'Paris rabbit got your back :)! yay!'

In [23]:
```
'1995'
```

Out[23]: '1995'

In [24]:
```
'doesn\'t'
```

Out[24]: "doesn't"

In [25]:
```
"doesn't"
```

Out[25]: "doesn't"

In [26]:
```
'"Yes," they said.'
```

Out[26]: '"Yes," they said.'

In [27]:
```
"\"Yes,\" they said."
```

Out[27]: '"Yes," they said.'

In [28]:
```
'"Isn\t," they said.'
```

Out[28]: '"Isn\t," they said.'

In [30]:
```
s = 'First line. \nSecond line.'
s
```

Out[30]: 'First line. \nSecond line.'

In [31]:
```
print(s)
```

```
First line.
Second line.
```

In [33]:
```
print('C:\some\ name')
```

```
C:\some\ name
```

In [34]:
```
print(r'C:\some\name')
```

```
C:\some\name
```

In [35]:
```
print("""\
      Usage: thingy [OPTIONS]
      -h                      Display this usage message
      -H                      Hostname to connect to
      """)
```

```
     Usage: thingy [OPTIONS]
     -h                      Display this usage message
     -H                      Hostname to connect to
```

In [36]:
```
3 * 'un' + 'ium'
```

Out[36]:
```
'unununium'
```

In [37]:
```
'Py' 'thon'
```

Out[37]:
```
'Python'
```

In [38]:
```
text = ('Put several string within parentheses'
        'to have them joinned together.')
```

In [39]:
```
text
```

Out[39]:
```
'Put several string within parenthesesto have them joinned together.'
```

In [45]:
```
prefix = 'Py'
result = prefix + 'thon'
print(result)   # Python
```

```
Python
```

In [46]:
```
prefix+'thon'
```

Out[46]:
```
'Python'
```

In [47]:
```
word = 'python'
```

In [48]:
```
word[0]
```

Out[48]:
```
'p'
```

In [49]:
```
word[5]
```

Out[49]:
```
'n'
```

In [50]:
```
word[-1]
```
Out[50]:
```
'n'
```

In [51]:
```
word[-2]
```
Out[51]:
```
'o'
```

In [52]:
```
word[-6]
```
Out[52]:
```
'p'
```

In [53]:
```
word[0:2]
```
Out[53]:
```
'py'
```

In [55]:
```
word[2:5]
```
Out[55]:
```
'tho'
```

In [56]:
```
word[:2]
```
Out[56]:
```
'py'
```

In [57]:
```
word[4:]
```
Out[57]:
```
'on'
```

In [58]:
```
word[-2:]
```
Out[58]:
```
'on'
```

In [59]:
```
word[:2] + word[2:]
```
Out[59]:
```
'python'
```

In [60]:
```
word[:4] + word[4:]
```
Out[60]:
```
'python'
```

In [61]:
```
word[42]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[61], line 1
----> 1 word[42]

IndexError: string index out of range
```

In [62]:
```
word[4:42]
```
Out[62]:
```
'on'
```

In [63]:
```
word[42:]
```
Out[63]:
```
''
```

In [64]:
```
word[0] = 'J'
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[64], line 1
----> 1 word[0] = 'J'

TypeError: 'str' object does not support item assignment
```

In [65]:
```
'J' + word[1:]
```

Out[65]:
```
'Jython'
```

In [66]:
```
word[:2] + 'py'
```

Out[66]:
```
'pypy'
```

In [67]:
```
s = 'supercalifragilisticexpialidocious'
```

In [68]:
```
len(s)
```

Out[68]:
```
34
```

In [69]:
```
squares = [1, 4, 9, 16, 25]
```

In [70]:
```
squares
```

Out[70]:
```
[1, 4, 9, 16, 25]
```

In [71]:
```
squares[0]
```

Out[71]:
```
1
```

In [72]:
```
squares[-1]
```

Out[72]:
```
25
```

In [73]:
```
squares[-3:]
```

Out[73]:
```
[9, 16, 25]
```

In [74]:
```
squares + [36, 49, 64, 81, 100]
```

Out[74]:
```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

In [81]:
```
cubes = [1, 8, 27, 65, 125]
4 ** 3
```

Out[81]:
```
64
```

In [82]:
```
cubes[3] = 64
```

In [83]:
```
cubes
```

Out[83]:
```
[1, 8, 27, 64, 125]
```

In [99]:
```
cubes.append(216)
cubes.append(7 ** 3)
cubes
```

Out[99]:
```
[1, 8, 27, 64, 125, 216, 343]
```

In [101…
```
rgb = ["Red", "Green", "Blue"]
rgba = rgb
id(rgb) == id(rgba)
```

Out[101…
```
True
```

In [102…
```
rgba.append("Alph")
rgb
```

Out[102…
```
['Red', 'Green', 'Blue', 'Alph']
```

In [103…
```
correct_rgba = rgba[:]
correct_rgba[-1] = "Alpha"
correct_rgba
```

Out[103…
```
['Red', 'Green', 'Blue', 'Alpha']
```

In [104…
```
rgba
```

Out[104…
```
['Red', 'Green', 'Blue', 'Alph']
```

In [106…
```
letters = ['a','b','c','d','e','f','g']
letters
```

Out[106…
```
['a', 'b', 'c', 'd', 'e', 'f', 'g']
```

In [108…
```
letters[2:5] = ['C', 'D', 'E']
letters
```

Out[108…
```
['a', 'b', 'C', 'D', 'E', 'f', 'g']
```

In [109…
```
letters[2:5] = []
letters
```

Out[109…
```
['a', 'b', 'f', 'g']
```

In [111…
```
letters[:] = []
letters
```

Out[111…
```
[]
```

In [112…
```
letters = ['a', 'b', 'c', 'd']
len(letters)
```

Out[112…
```
4
```

In [113…
```
a = ['a', 'b', 'c']
n = [1, 2, 3]
x = [a, n]
x
```

Out[113… [['a', 'b', 'c'], [1, 2, 3]]

In [114…
```
x[0]
```

Out[114… ['a', 'b', 'c']

In [115…
```
x[0][1]
```

Out[115… 'b'

In [116…
```
a, b = 0, 1
while a < 10:
    print(a)
    a, b = b, a+b
```

```
0
1
1
2
3
5
8
```

In [117…
```
i = 256*256
print('The value of i is', i)
```

```
The value of i is 65536
```

In [ ]:
```
a,b = 0, 1
while a < 1000:
    print(a, end=',')
```